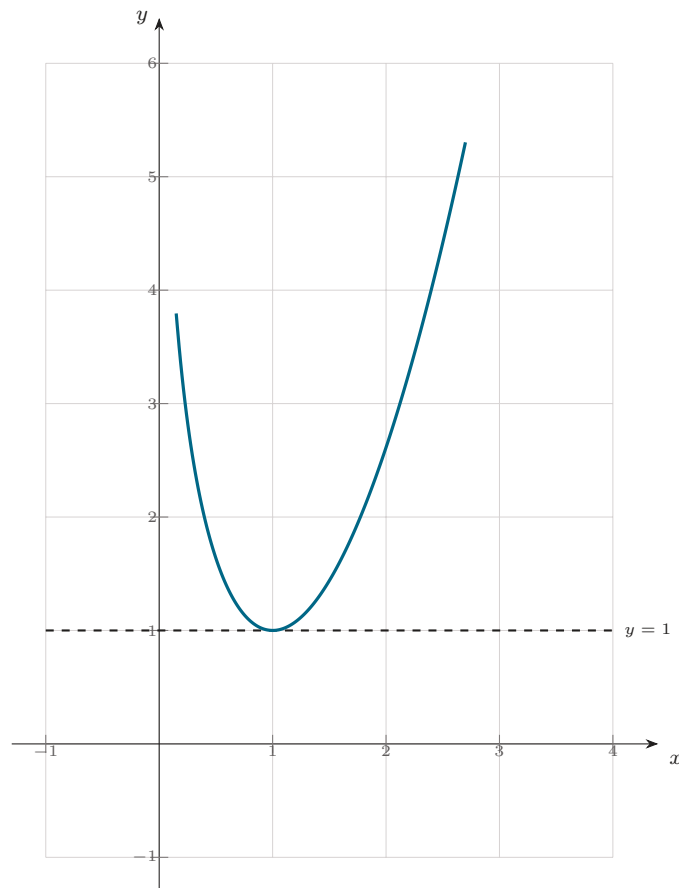


Corrigé — Exercice 10

1) **a)** $\lim_{0+} f = +\infty$ (car $-2 \ln x \rightarrow +\infty$) : asymptote verticale $x = 0$. **b)** $\lim_{+\infty} f = +\infty$. **2)** **a)** $f'(x) = 2x - \frac{2}{x} = \frac{2(x^2 - 1)}{x}$. **b)** $f' < 0$ sur $]0, 1[$, > 0 sur $]1, +\infty[$: minimum $f(1) = 1$. **3)** Le minimum vaut 1, donc $f(x) \geq 1$. **4)** **a)** $f''(x) = 2 + \frac{2}{x^2} > 0$: $\mathcal{C}_f$ convexe. **b)** $f'(1) = 0$, $f(1) = 1$: $y = 1$. **5)** **a)** $F'(x) = x^2 - 2(\ln x + 1) + 2 = x^2 - 2 \ln x = f$. **b)** $\int_1^e f = [F]_1^e = \frac{e^3 - 7}{3}$.



$\mathcal{C}_f$ et la tangente $y = 1$ (Exercice 10).